

TEST PLAN

Digital Twin Test Plan

Team Name:	ROFASO
Project Name:	Damage Assessment Protocol (D.A.P.)
Part / System Name:	Digital Twin
Prepared By:	Jordan Cummings, Henry Hong, Letizia Laura

Approval

Name / Role	Signature	Date

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1. Introduction

1.1. Purpose

The following plan aims to test the ability of the Digital Twin (DT) to take in seaglider mission data files, compile desired input parameters into a matrix form and output the appropriate simulated data matrix. The DT must be able to simulate expected behaviors for nominal and off-nominal Seaglider geometries, including broken wing, broken rudder and no-damage cases. To test this dives from the 4/17/2026 deployment in Edmonds will be fed to the D.T and it's outputs will be compared to the corresponding mission data for accuracy. The requirements being verified in this test procedure are the following:

- DTR-1: The Digital Twin shall output orientation (pitch angle, roll angle, and heading) with average angular error no greater than 19.8% relative to the maximum change in angle from onboard-recorded data.
- DTR-3: The Digital Twin shall output vertical position with percent error no greater than 6.6% of the maximum mission depth relative to onboard-recorded depth.
- DTR-4: The Digital Twin shall output vertical velocity with percent error no greater than 3.3% of the maximum change in vertical velocity relative to onboard-recorded vertical velocity.

2. Test Variables

Table 1: Test Variables

Variable	Description	Values
Top Rudder	Percentage of rudder mounted on the Seaglider for a given dive	100%, 0%
Port Wing	Percentage of wing mounted on the Seaglider for a given dive	100%, 25%

3. Test Scenarios and Damage Modes

3.1. Nominal Configurations (No Damage Modes/Fault Cases)

1. FLT0: 100% of wing and 100% of rudder attached

3.2. Off Nominal Configurations (Damage Modes/Fault Cases)

1. FLT1: 50% of starboard wing span remaining, 100% of top rudder span remaining
2. FLT2: 100% of starboard wing span remaining, 50% of top rudder span remaining

Table 2: Fault Test Needs Matrix

	% Span Remaining	Starboard Wing			
		0%	25%	50%	100%
Top Rudder	0%				
	25%				
	50%				FLT2
	100%			FLT1	FLT0

4. Test Matrix

4.1. Test Run Matrix

Test ID	Edmonds 4/17 .nc/.log File Name	ϕ Pass?	θ Pass?	Heading Pass?	Depth Pass?	Velocity Pass?
FLT0	p1940005.log p1940005.nc					
FLT1	p1940008.log p1940008.nc					
FLT2	p19400011.log p19400011.nc					

5. Success Criteria

Requirement	Validation Method	Pass Criteria
DTR-1: Orientation Outputs	Plotting Digital Twin versus Mission data and verify the two datasets are within 19.8% difference	The graphs for ϕ , θ and heading are within their respective error bounds
DTR-3: Vertical Position Outputs	Plotting Digital Twin versus Mission data and verify the two datasets are within 6.6% difference	Depth is within its respective error bound
DTR-4: Vertical Velocity	Plotting Digital Twin versus Mission data and verify the two datasets are within 3.3% difference	Vertical Velocity is within its respective error bound